

----- Active space -----

NEONATAL RESUSCITATION

Priority of management in newborn :

1st : **Respiration** (Crying after birth).

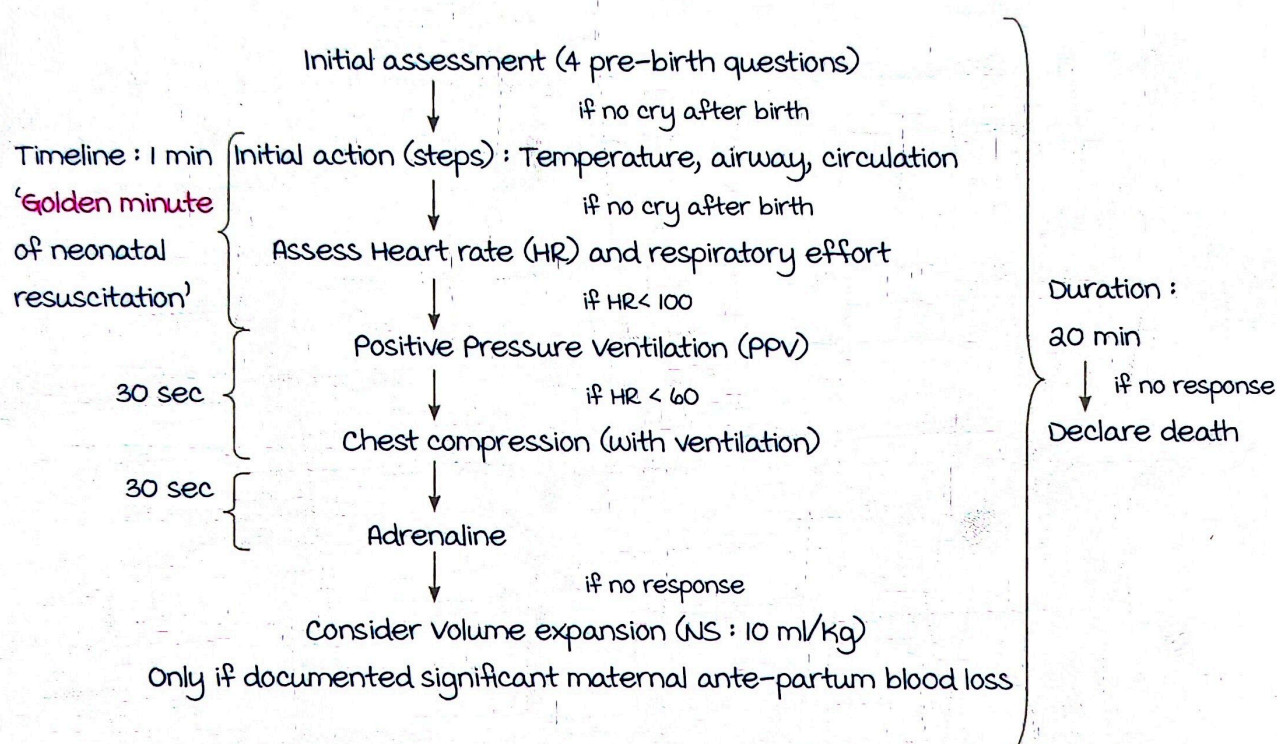
2nd : Temperature.

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3rd : Circulation.

Management Algorithm

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INITIAL ASSESSMENT :

4 pre-birth questions : To predict and plan for outcome of delivery.

1. Term gestation ?
2. Clear amniotic fluid ?
3. Risk factors present ? (maternal or fetal).
4. Umbilical cord management plan :
 - Delay cord clamping for 30-60s after birth ? (Only if baby cried after birth).
 - Cord blood for newborn screening taken ?

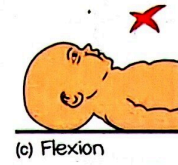
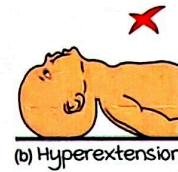
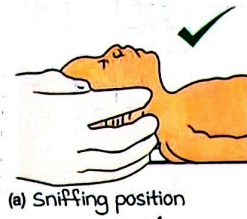
INITIAL STEPS :

1. Temperature : Keep baby under warmer.

2. Breathing : Gentle tactile stimulation

• methods :

- Rubbing over back (with cotton towel)/
- Flicking soles.



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3. Airway :

- Positioning : Slight extension of neck (Sniffing position) using towel under shoulder.

- Suctioning : Only if visible secretions.

Order : **mouth** → **Nose**.

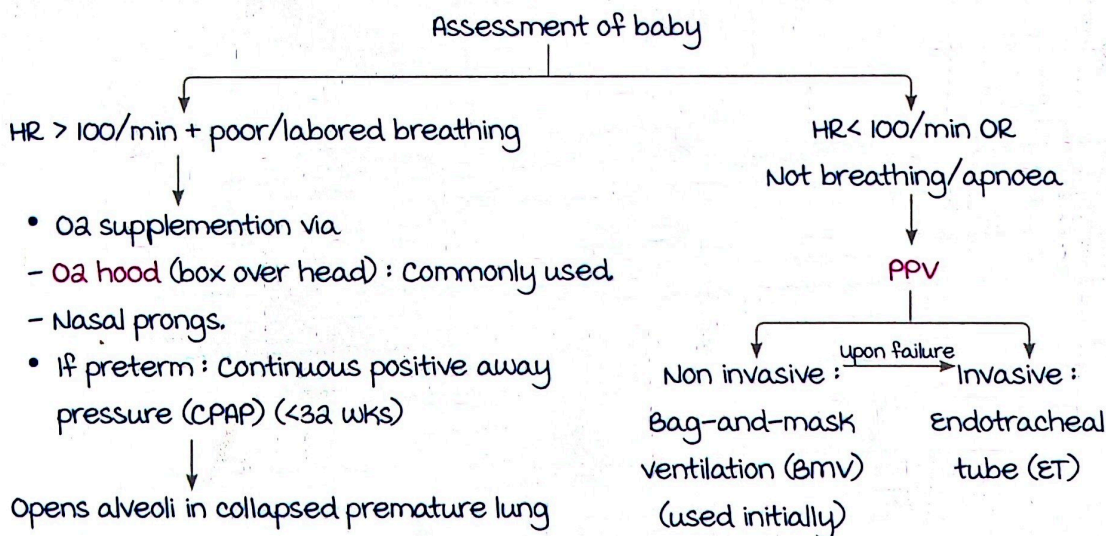
(Suctioning nose first → blocking of nose → mouth breathing → Secretions aspirated).

ASSESSMENT OF HR AND RESPIRATORY EFFORT :

Purpose of HR assessment :

No cry after birth (No breathing) → **significant hypoxia** → ↓ HR (< 100/min).

Note : Normal newborn HR : > 100/min.

**BAG AND MASK VENTILATION :**

Volume : 240 to 750 ml.

Rate : 40-60/min.

Response monitoring : HR > 100/min.

Pressure { 1st pressure : 30-40 cm of H₂O (Higher to open stiff airway).
 Subsequent : 15-20 cm of H₂O (Lower to prevent hyper inflation/rupture of alveoli).

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FIO₂ : (Based on gestational age).

- ≥ 35 weeks room air (21% O₂).
- < 35 weeks start with 21-30% O₂ → Titrate to target SpO₂ (measured at right hand).

mins after birth	Targeted SpO ₂
1 min	60-65%
2 min	65-70%
3 min	70-75%
4 min	75-80%
5 min	80-85%
10 min	85-95%

Need for matching targeted SpO₂:Excess O₂, (Preterm)

Hyperoxia damage :

Retinopathy of prematurity (ROP) :



Blindness

Ventilation correction steps if response inadequate : In order undertaken :

mnemonic : **MRSOPA**.1. **mask** repositioning :- Incorrect position : **m/c** cause of inadequate response.

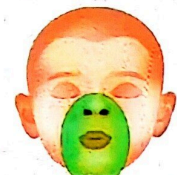
Correct



Incorrect closure



Too small



Too large

2. **Readjust** head position.3. **Suction** : Remove all secretions.6. **Alternate** airway : If all steps fail.

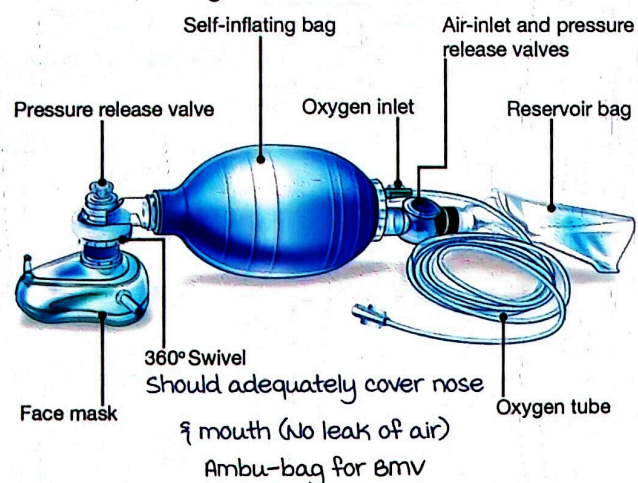
(ET tube)

4. **Open** mouth.5. **Pressure** increase over bag.

Absolute C/I : Congenital diaphragmatic hernia.

Air enters via esophagus → Inflates intrathoracic viscera → Compression of lungs (↑ respiratory distress).

Goal : Respiratory rate 40 to 60 breaths/min



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Endotracheal intubation :

visualization of airway : via straight blade laryngoscope

Size 0 : Preterm	} on back of blade
Size 1 : Term.	

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Type of ET : **Uncuffed**

- To prevent **pressure tracheal necrosis** (D/t cuff pressure on narrow sub-glottic area).

Inner diameter of tube :

- Based on weight of body :
 - $< 1 \text{ kg}$: 2.5 mm.
 - 1 to 2 kg : 3 mm.
 - $> 2.5 \text{ kg}$: 3.5 mm.

End tidal CO₂ estimation (EtCO₂) : **Best method** to determine if placement is correct done using capnography

Insertion depth : Naso-tragal length (in cm) + 1 cm.

CHEST COMPRESSION :

Done alongside ventilation (2 person).

Ratio : **3 : 1** (90 compressions : 30 ventilation in 1 min).

Response monitoring : using cardiac monitor.

- HR > 60 .

ADRENALINE :

Route :

- Preferred : IV (Through umbilical vein).

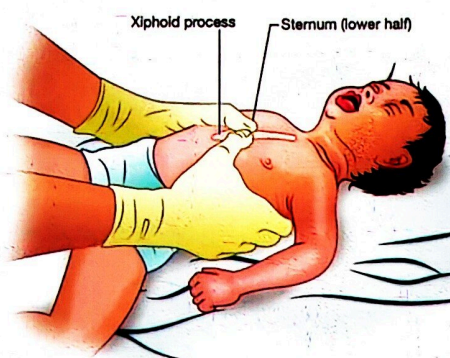
↓ if inaccessible

Intraosseous route

Dose : **0.2 ml/kg**.

Strength : 1 : **10,000**.

Lower 1/3rd
Depth : 1/3rd AP diameter



2-thumb technique
performed from head end usually